

EDUGENIE AI

Project Design, Planning, Solution Architecture and Performance Testing

Project Name	EduGenie AI
Project Type	Generative AI Educational Assistant
Document	Project Design and Performance Testing

Prepared for the EduGenie AI project

1. Initial Project Planning

Product Backlog, Sprint Schedule, and Estimation

The following project planning content is adapted to the EduGenie AI project while preserving the structure and user-story wording shown in the supplied template.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Registration	USN-1	As a user, I can register for the application by entering my email and password, and confirming my password.	2	High			
Sprint-1		USN-2	As a user, I will receive a confirmation email once I have registered for the application	1	High			
Sprint-2		USN-3	As a user, I can register for the application through Facebook	2	Low			
Sprint-1		USN-4	As a user, I can register for the application through Gmail	2	Medium			
Sprint-1	Login	USN-5	As a user, I can log into the application by entering email & password	1	High			

Note: Date, Team ID / Group Number, maximum marks, and external organization branding have been intentionally removed as requested.

2. Problem-Solution Fit

EduGenie AI – Problem-Solution Fit Canvas

This canvas follows the supplied template structure and is organized specifically around the EduGenie AI educational assistant.

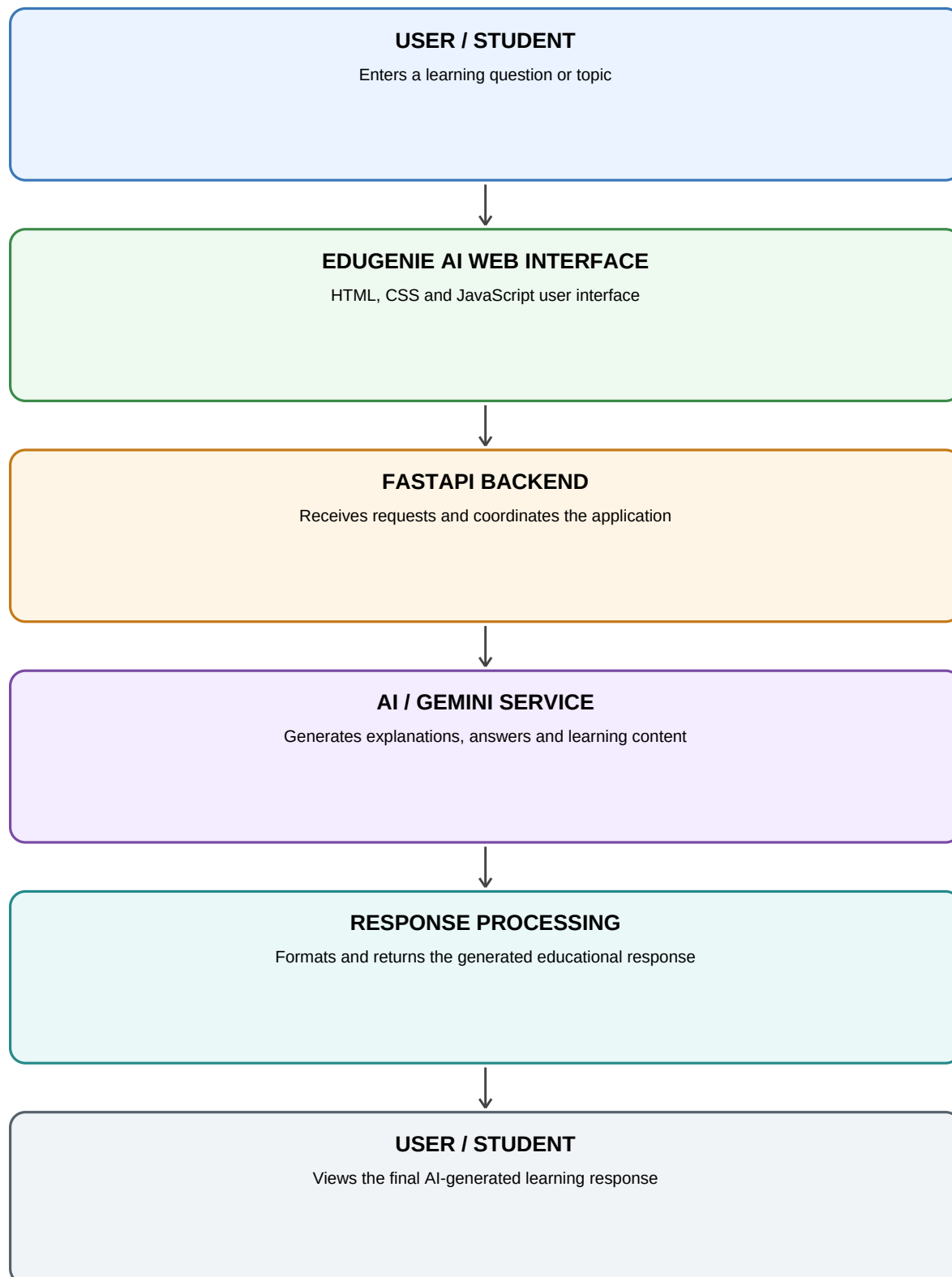
1. CUSTOMER SEGMENT(S) [CS] Students, learners and users seeking quick educational support.	6. CUSTOMER LIMITATIONS Limited study time, difficulty finding concise explanations, varying learning levels and limited access to personalized guidance.	5. AVAILABLE SOLUTIONS Search engines, textbooks, online learning platforms and general-purpose AI assistants.
2. PROBLEMS / PAINS + ITS FREQUENCY [PRI] Difficulty understanding topics, finding relevant explanations and getting quick answers.	9. PROBLEM ROOT / CAUSE [RC] Information is spread across multiple sources and explanations may not match the learner's level or immediate need.	7. BEHAVIOR + ITS INTENSITY [BE] Users repeatedly search, compare sources and ask for simpler explanations.
3. TRIGGERS TO ACT [TR] A question, difficult topic, assignment, revision need or request for an explanation.	10. YOUR SOLUTION [SL] EduGenie AI provides an AI-powered educational interface that accepts a learning query and generates a clear, contextual response.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Web application / browser OFFLINE Prepared learning material and saved outputs where supported.
4. EMOTIONS BEFORE / AFTER [EM] Before: confusion or uncertainty. After: clearer understanding and a structured response.		

3. Solution Architecture

EduGenie AI – High-Level Architecture

The architecture maps the presentation layer, application/API layer, AI service and data/storage responsibilities for EduGenie AI.

EduGenie AI – System Flow Diagram



4. Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where learners enter questions and view generated educational responses.	HTML, CSS, JavaScript
API Gateway / Backend	Receives client requests, validates input, coordinates application logic and returns responses.	FastAPI, Python
AI / Core Logic Service	Processes the learning request and communicates with the generative AI model to produce educational content.	Python, Google Gemini API
Response Processing	Formats the generated response and sends it back to the web interface for display.	Python, FastAPI
Data / Storage	Stores application data or generated resources when storage is required by the implementation.	Project storage / file system

Architecture Flow

User / Student → EduGenie AI Web Interface → FastAPI Backend → AI / Gemini Service → Response Processing → User / Student

5. Performance Testing

Performance Testing Overview

Performance testing evaluates how the EduGenie AI application behaves under expected and peak load conditions. The following sections document the testing approach, scenarios and results.

Field	Details
Testing Tool Used	JMeter / Locust / Postman (as applicable)
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	EduGenie AI web application / API endpoint used for response generation
Test Environment	Local / Cloud / Hosted Server (as applicable)
Test Date	To be filled after the test is executed

Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single-user educational query generation	1	60	Successful response without errors
2	Concurrent users sending educational queries	10	60	Stable responses under normal load
3	Higher concurrent request load	25	120	Application remains responsive
4	Short-duration spike in requests	50	30	Graceful handling of increased traffic

6. Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	To be measured	Pending	Record measured value
2	Response Time (Max)	< 5 seconds	To be measured	Pending	Record measured value
3	Throughput (Req/sec)	Project target	To be measured	Pending	Record measured value
4	Error Rate	< 1%	To be measured	Pending	Record measured value
5	CPU Utilization	< 80%	To be measured	Pending	Record measured value
6	Memory Utilization	< 80%	To be measured	Pending	Record measured value

Observations & Analysis

Key Findings	Describe what the test results reveal about the EduGenie AI system's performance. Actual findings should be entered after testing.
Bottlenecks Identified	List any performance bottlenecks observed during testing, such as API latency, high CPU usage or slow response generation.
Optimization Steps Taken	Describe changes or optimizations made to improve performance, if any.

Screenshots / Evidence

Insert performance-testing screenshot here

Insert performance-testing screenshot here